

DeepSkinNet: A Deep Learning Model for Skin Cancer Detection

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Abstract—In this article, an intelligent skin cancer detection system is proposed. It is common knowledge that skin cancer is one of the deadliest diseases in the world. Therefore, in order to save human lives, it must be detected correctly at the early stage, for which a fully automatic system using Deep Learning techniques can be used. In this work, the HAM10000 dataset is used for skin cancer image classification. A specific model for skin cancer classification, DeepSkinNet model, is proposed and tested. For skin lesion classification, the proposed DeepSkinNet model is compared to contemporary models such as AlexNet, VGG-16, and InceptionV3. The same dataset is trained with all the above models and the confusion matrix of the models is determined. Known quantitative measures such as accuracy, precision and recall are also evaluated. Finally, the experimental studies shows that the proposed model achieves better accuracy than the other models. The DeepSkinNet model has an accuracy, precision, recall and F1-score of 97.354%, 98%, 97% and 97% respectively.

Index Terms—DeepSkinNet model, deep learning, skin lesions, skin cancer detection

I. INTRODUCTION

The skin is our largest organ, providing sensations such as touch, warmth, cold, etc. The skin's primary job is to shield the human body from external factors such as germs, toxins, and temperature. The skin regulates the temperature of the human body. Skin lesions can occur after delivery or due to improper skin care. They are dangerous if they turn out to be cancerous. There are different types of skin cancer. This article deals with melanocytic nevi, melanomas, benign keratoses, basal cell carcinomas, actinic keratoses, vascular lesions and dermatofibromas. Melanoma is the worst skin cancer because it spreads rapidly and can even lead to death. As a result, early detection of skin cancer is critical. The most difficult thing is that skin lesions are hard to see. Today, there are several dermatoscopic imaging techniques that can capture images of skin lesions. Recently, several investigators have successfully employed deep learning and transfer learning techniques to classify objects, which can also be employed to classify skin lesions [1].

The objective of the reported work is to find a better solution to the challenges of skin cancer detection. The focus is to develop a skin cancer detection algorithm on RGB images in the HAM10000 dataset. In this work, we have developed a DeepSkinNet model from scratch using convolutional neural

network for skin lesion classification and compared its performance with the known pre-trained models to find the best model for skin lesion classification. The system improves the classification accuracy and reduces the loss. The performance of the proposed system is analysed in terms of classification accuracy, loss and confusion matrix.

A. Organization of the Paper

The paper is structured as follows. Section II discusses closely related work. Section III discusses background information on this work. Section IV outlines the proposed technique of the method. Results and discussion can be found in Section V. Section VI summarises the work.

II. LITERATURE REVIEW

This section provides an overview of the most related techniques for detecting skin cancer using Deep Learning. The main challenge in detection is that the images of different skin cancers look similar, so their classification becomes complicated. A hybrid model for the categorization of melanomas has been suggested by Codella et al. This model makes use of SVM, sparse coding, and Deep Learning approaches, and it obtains an accuracy rate of 93% [2]. There are also some other great works on skin cancer classification. Ozkan and Koklu developed a classification system for skin lesions [3]. They classified skin lesions into melanoma, abnormal and normal skin lesions using four different models. Among the four models, the artificial neural network achieved a better accuracy of about 92%. Chakravorty et al [4] proposed a classification of skin lesions considering the irregular colour distribution in the skin lesion using Kullback-Liebler divergence and structural similarity matrix and achieved an accuracy of 83%. Using multiple scale-based lesion representation and joint reverse classification, Bi et al. developed a system for classifying skin lesions into melanoma and non-melanoma categories, with a prediction accuracy of 92% [5]. Deep and transfer learning are utilised in the process of constructing a skin lesion classifier outlined in [6]. The model provided was both trained and tested using the ph2 dataset. [7] demonstrated a hybrid strategy for detecting melanomas utilising Deep Learning and traditional machine learning techniques. They used a set of data indicating the edges, texture, and colour of a skin lesion

to train a CNN as well as two machine learning classifiers. To improve the performance of the model, a majority decision is made by combining the results of the models. The experiment shows that a combination of the three models gives the highest accuracy. It was reported in [8] that Deep Learning was used to construct a skin cancer classification system for pigmented skin lesions. This system was used to diagnose skin cancer.

In an attempt to increase both the speed at which skin cancer may be identified and the precision with which it can be detected, researchers have developed algorithms for autonomous skin cancer detection. The problem, however, was that they did not have enough data for classification. So the accuracy was low and since only a small amount of data was available, the prediction could be wrong and therefore they are not reliable. To solve these problems, we employed the HAM10000 dataset, that consists of different skin lesions. DeepSkinNet is a modified CNN model that is proposed in this study for the purpose of skin cancer classification. The performance of DeepSkinNet is evaluated with the help of state-of-the-art benchmarks.

III. BACKGROUND OF WORK

A. AlexNet

AlexNet is comprised of eight layers, each with their own set of learnable parameters [9]. The Fig. 1 shows the schematic diagram of AlexNet 1. The model is made up of a total of five convolutional layers, one of which combines max-pooling with other operations, and is then followed by three fully connected layers [10]. All layers use Relu as the activation function, except for the output layer. The model takes an image of size $28 \times 28 \times 3$ as input. The convolution layer will have a size of 11×11 with 96 filters and stride four. "Relu" activation is used in this layer. There are basically three fully connected layers in which the last layer act as an output layer. The output layer will have seven output neurons because here we are classifying the dataset into seven classes. At the output, a dropout layer is added at the rate of 0.5. There are 63.7 million parameters in an AlexNet model. To achieve better results in AlexNet, the hyperparameters of the pre-trained model were adjusted and the accuracy of the model was increased.

B. VGG-16

VGG-16 is a sixteen-layer convolutional neural network architecture with thirteen convolutional layers, five max-pooling layers, and three dense layers, for a total of 21 layers but only sixteen weighting layers [11]. The schematic diagram of VGG-16 architecture is shown in Fig. 2. The input tensor size for VGG-16 is 224×224 with three RGB channels. The significant feature of VGG-16 is that instead of focusing on a huge number of hyperparameters, they focused on convolution layers with a 3×3 filter with Stride 1 and always used the same padding and maxpooling layer with 2×2 filters with Stride 2. The convolution layer and maxpool layer are arranged similarly throughout the design. Conv-1 has 64 filters, Conv-2 has 128, Conv-3 has 256, and Conv-4 and Conv-5 have 512. After a stack of convolutional layers, three Fully-Connected

(FC) layers are added: the first two have 4096 channels each, while the third performs a classification into seven classes. The Soft Max layer is the dense output layer.

C. InceptionV3

The Inception Net is a substantial advancement in CNN classifiers. Inception Net performed the best on ImageNet. The deep neural Inception network replicates the Inception modules to achieve the required depth [12]. The Fig. 3 shows the basic architecture of the InceptionV3 model. Inception uses three convolutional layers in addition to a max-pooling layer. The convolutional layers have filters of different sizes. The module concatenates all filter maps. The model learns from several parallel filters of of varying scales and sizes. Fig. 3 shows the schematic diagram of Inception module. Within a single network module, Inception is capable of functioning as a "multi-level feature extractor" by carrying out convolutions with a factor of 1×1 , 3×3 , and 5×5 , respectively. Inception comes in several variants. InceptionV3 is an improved version used in biomedical applications. Its 48 layers include 24 million transfer learning parameters.

IV. PROPOSED METHODOLOGY

We have developed a DeepSkinNet model specifically for skin cancer detection. The main advantage of the proposed DeepSkinNet model is that it requires very little preprocessing compared to other algorithms. The proposed model architecture is depicted in Fig. 4. It has a combination of local or global pooling layers. The pooling layers are mainly used for dimensionality reduction of the data. There are many types of pooling such as max-pooling, average-pooling and sum-pooling. We employed max-pooling for dimensionality reduction. Fully connected layers are used to connect a neuron in one layer to another layer. In a fully connected layer, a traditional multilayer perceptron neural network (MLP) is used. After the image passes through the flattened layer, it is converted into a 1D array and then forwarded to the fully connected layer to classify the images. The proposed DeepSkinNet model used for this work takes the images as RGB images. Here we have used the HAM10000 dataset which consists of RGB images of different skin lesions. The proposed model consists of 7 learnable layers which take the skin lesion image as input. There are 4 convolutional layers with max-pooling layers. It has 2 fully connected layers and one output layer. The entire model uses "Relu" as the activation function except for the output layer. The output layer uses "Softmax" as the activation function. This model is dedicated to skin cancer detection. The above architecture contains a total of 13.4 million parameters.

V. RESULTS AND DISCUSSIONS

A. Dataset and Experimental Setup

The HAM10000 dataset was used for performance analysis of the proposed method with skin cancer images. The dataset consists of 10015 dermatoscopic images of seven different categories. These include melanocytic nevi (nv), melanomas

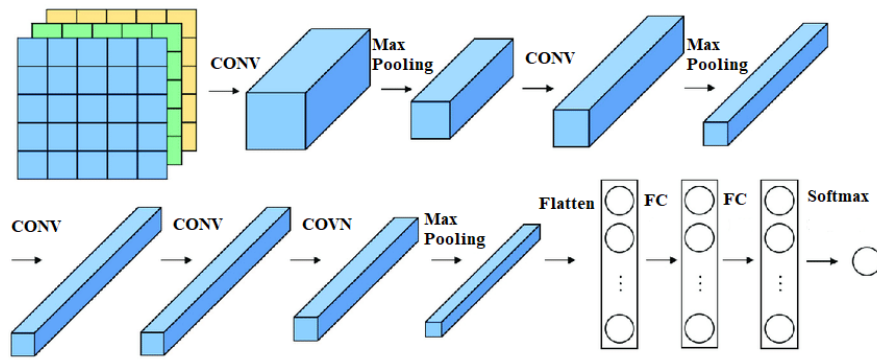


Fig. 1: Schematic diagram of AlexNet

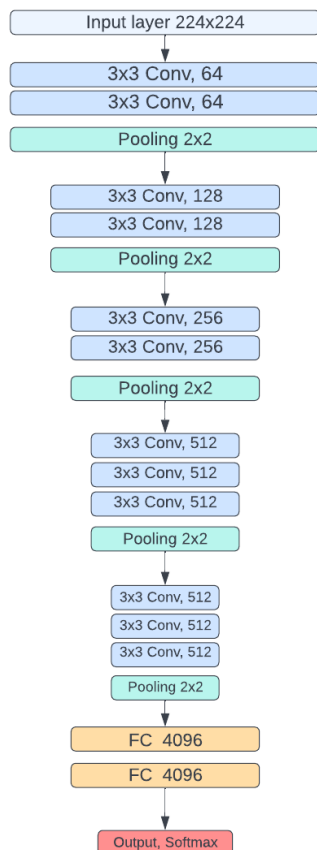


Fig. 2: Schematic diagram of VGG-16

(mel), benign keratoses (bkl), basal cell carcinomas (bcc), actinic keratoses (akiec), vascular lesions (vasc) and dermatofibromas (df). Table I show the the number of images in different classes of the dataset. The main problem with the dataset is that there is an imbalance between the different classes. To solve this problem, the dataset is expanded by data augmentation to 45756 and the images in each class are balanced to achieve a better result and avoid misclassification. Here the data is expanded by rotation, width shift, height shift,

TABLE I: Images in different classes of the dataset

Categories	Number of images
Melanocytic nevi	6705 images
Melanoma	1113 images
Benign keratosis	1099 images
Basal cell carcinoma	514 images
Actinic keratoses	327 images
Vascular lesions	142 images
Dermatofibroma	115 images

TABLE II: Classification summary of DeepSkinNet model

Classes	Precision	Recall	F1 Score	Support
Melanocytic nevi	1.00	0.96	0.98	1374
Melanoma	0.89	1.00	0.94	205
Benign keratosis	0.92	1.00	0.96	227
Basal cell carcinoma	0.97	1.00	0.98	94
Actinic keratoses	0.95	1.00	0.97	55
Vascular lesions	1.00	1.00	1.00	28
Dermatofibroma	0.95	1.00	0.98	20

horizontal flipping and vertical flipping to balance the data. This is done using a package in Python called "Image Data Generator", a very useful tool in Python for data management. After solving this problem, the size of the input image is resized into $28 \times 28 \times 3$. Then the image dataset is converted into corresponding NumPy arrays for faster detection.

The experiment is conducted on an Intel i5 processor with 8 GB RAM. Jupyter Notebook is the platform used for skin cancer detection. All the coding is done on the above experimental step. In the work, we have used the optimizer "NAdam" optimizer for optimisation, the loss function used for the work is "sparse categorical cross-entropy". We have

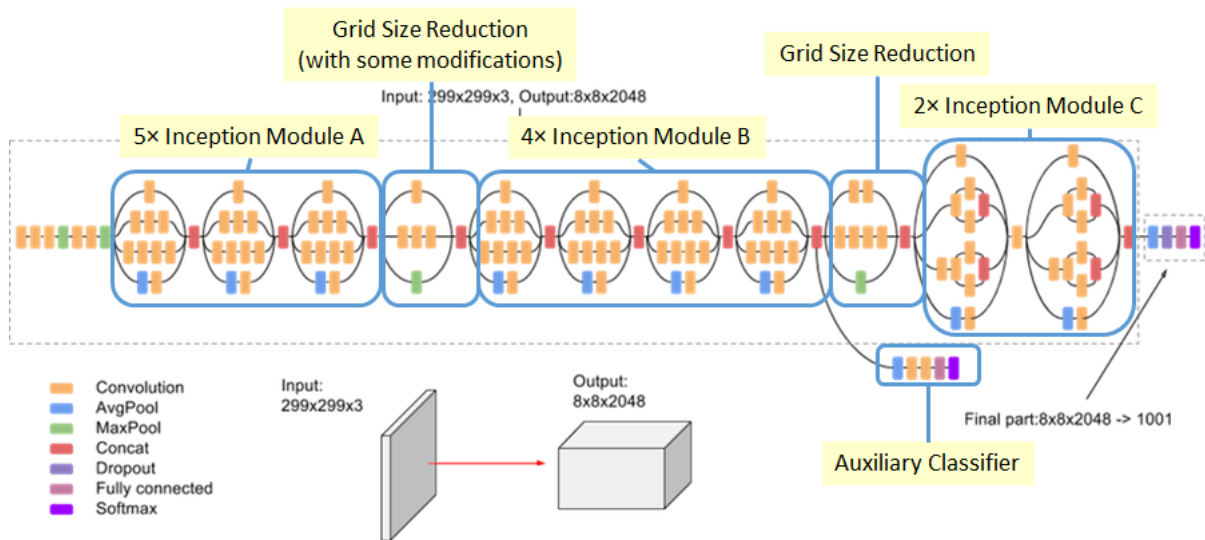


Fig. 3: Schematic diagram of InceptionV3

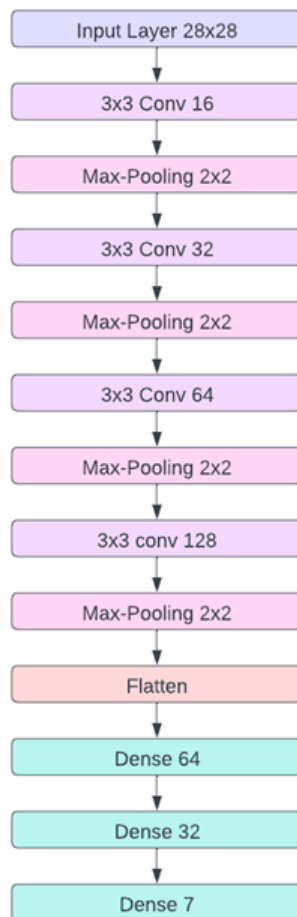


Fig. 4: Schematic diagram of DeepSkinNet model

TABLE III: Classification summary of AlexNet

Classes	Precision	Recall	F1 Score	Support
Melanocytic nevi	1.00	0.95	0.98	1374
Melanoma	0.89	0.99	0.94	205
Benign keratosis	0.87	1.00	0.93	227
Basal cell carcinoma	0.95	1.00	0.97	94
Actinic keratoses	0.95	1.00	0.97	55
Vascular lesions	0.97	1.00	0.98	28
Dermatofibroma	0.95	1.00	0.98	20

rate when there is no improvement in accuracy. Early stopping is also applied to stop training when there is no improvement in the accuracy. The batch size used for the work is 64, the validation split is 0.2 and the epochs are fixed at 50. Performance is analyzed in terms of accuracy, precision, recall, and F1 score.

B. Performance with the DeepSkinNet model

The DeepSkinNet model for skin cancer detection has been implemented and performance has been noted. The training accuracy of the model is 100% while the validation and testing accuracy of the model are 97.92% and 97.34% respectively. The precision and recall of the model are shown in the classification summary Table II. The accuracy curve, loss curve and confusion matrix for the model are shown in Fig. 5 and Fig. 6, respectively. Fig. 5 (i) shows the accuracy curve of the DeepSkinNet model. As we can see, the training accuracy starts at 0.32 and increases to about 1.0 at the fourteenth epoch and then remains stable. The validation curve starts

also employed “ReduceLROnPlateau” to reduce the learning

TABLE IV: Comparison of skin cancer detection on the HAM dataset with different models

Model	Accuracy	Loss	Precision	Recall	F1 Score
CNN(Skin Cancer detection model)with Nadam optimizer	0.9735	0.09	0.98	0.97	0.97
Alex Net with Nadam optimizer	0.96605	0.259	0.97	0.97	0.97
CNN(Skin Cancer detection model)with Adam Optimizer	0.97004	0.225	0.97	0.97	0.97
Alex Net with Adam optimizer	0.88717	0.293	0.92	0.89	0.90
VGG-16	0.8635	0.1001	0.8688	0.8603	0.8645
Inception V3	0.8625	0.3458	0.8625	0.8425	0.8524



Fig. 5: DeepSkinNet model:(i) Training and Validation Accuracy (ii) Training and Validation Loss

at 0.6 and increases to 0.97354 at the 16th epoch and then becomes constant. From the graph, it is clear that the model has a training accuracy of 1.0 and a validation accuracy of 0.97354. Fig. 5 (ii) shows the loss curve of the CNN model.

Fig. 6 shows the confusion matrix of the DeepSkinNet model. A total of 2003 images from different classes were taken for testing. 0- 'nv', 1- 'mel', 2- 'bkl', 3- 'bcc', 4- 'akiec', 5- 'vasc', 6- 'df'. These are the class labels for the different classes in the dataset. With the exception of class 0 and class 1, all classes are accurately predicted. For class 0, 1322 images were predicted correctly and the incorrect prediction of class 0 is shown in Fig. 6. Similarly, class 1 was also predicted incorrectly as shown in Fig. 6.

C. Performance with the state-of-the-art models

The skin cancer detection system based on the AlexNet model has been implemented and performance has been noted. The training accuracy of the model is 99.54% while the validation and testing accuracy of the model are 97.55% and 96.605% respectively. The precision and recall of the model

	0	1	2	3	4	5	6
0	1322	24	21	3	3	0	1
1	1	204	0	0	0	0	0
2	0	0	227	0	0	0	0
3	0	0	0	94	0	0	0
4	0	0	0	0	55	0	0
5	0	0	0	0	0	28	0
6	0	0	0	0	0	0	20
	0	1	2	3	4	5	6

Fig. 6: Confusion Matrix with DeepSkinNet model

are shown in the classification Table in Table III. The training accuracy starts at 0.55 and increases approximately to 0.989 at

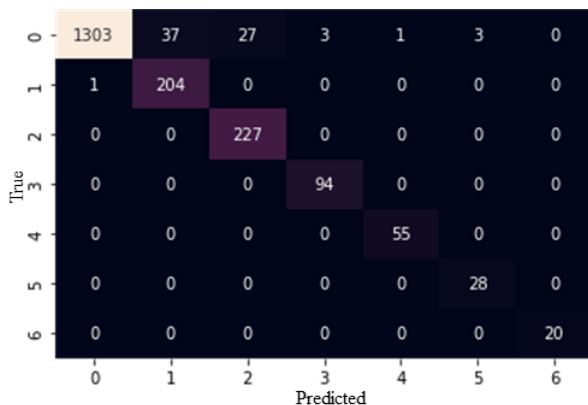


Fig. 7: Confusion Matrix with AlexNet

the fourteenth epoch and then remains stable. The validation curve starts at 0.29 and increases to 0.9604 at the sixteenth epoch and then becomes constant. The model achieves a training accuracy of 0.989 and a validation accuracy of 0.9604. The training loss starts at 1.25 and decreases to about 0.12 at the fourteenth epoch and then becomes stable. The validation curve starts at 2.45 and drops to 0.145 at the fourteenth epoch and then becomes constant. The model achieves a training loss of 0.12 and a validation loss of 0.145. Fig. 7 shows the confusion matrix of the AlexNet model. A total of 2003 images from different classes were taken for testing. These classes are predicted and the confusion matrix is plotted. All except class 0 and class 1 are predicted correctly. For class 0, 1303 images were predicted correctly and the wrong prediction of class 0 can be seen in Fig. 7. Similarly, class 1 was also predicted incorrectly as can be seen in Fig. 7.

The training accuracy of the VGG-16 model is 99.49% while the validation and testing accuracies of the model are 86.35% and 83.605% respectively. The training accuracy of the inceptionV3 model is 99.25% while the validation and test accuracy of the model are 86.25% and 79.605% respectively. The performance of the proposed DeepSinNet model compared to the state-of-the-art models is shown in Table IV. These results clearly show the superiority of the proposed DeepSkinNet model. The advantage of the proposed model is that it improves the prediction accuracy and speed compared to the other models used in this study. Moreover, the proposed DeepSkinNet model is lightweight. The experimental results show that skin cancer detection using Deep Learning techniques is promising and further research and work is needed to improve the reliability and efficiency of the model.

VI. CONCLUSIONS

Skin cancer is one of the deadliest diseases in the world and should be detected at an early stage. Therefore, the detection of skin cancer must be done very quickly, which has the advantage of automatic detection. In the initial studies, researchers used a small dataset for skin cancer classification. Until now, the biggest challenge in skin cancer detection has been the availability of a large, reliable dataset. This

task is solved by the HAM10000 dataset, which consists of 10015 images from 7 different classes. In this paper, DeepSkinNet - a skin cancer classification model - is proposed and evaluated. For skin lesion classification, the proposed DeepSkinNet model is compared with current models such as AlexNet, VGG-16 and InceptionV3. The performance of the models is analysed in terms of accuracy, precision, detection and confusion matrix. The training, validation and testing accuracies of the proposed DeepSkinNet model are 100%, 97.92% and 97.34% respectively, which are superior to all other deep learning models considered in this study.

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